

Shapes, forces, and torques of compressed elastic fluid interfaces: beyond axisymmetric configurations

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Abstract

Inspired by the classical Plateau–Douglas problem for soap films bounded by two closed curves, we solve an analogous problem for fluid interfaces with fixed surface area and resistance to out-of-plane bending. The boundaries are planar but need not be symmetric or concentric. The equilibrium surfaces minimize the Willmore bending energy and generalize minimal surfaces such as the catenoid while also exhibiting characteristic features of confined elastic interfaces such as buckling. We systematically classify all possible buckling modes and solution branches by performing a weakly nonlinear analysis and developing a fully nonlinear spectral solver. Our mathematical framework provides insight into the forces and torques required to stabilize cellular membranes and other soft materials and shows that physically relevant asymmetric states can arise even in symmetric systems.

1 Introduction

We study the mechanics of topologically cylindrical fluid interfaces with a resistance to out-of-plane bending that are attached to two closed boundaries. Such configurations occur naturally during cellular processes such as scission and endo- and exocytosis [15, 7]. They also appear *in vitro* in mechanical tests, in which instruments such as cantilevers, optical tweezers, or pipettes are used to deform a microscopic object, and the resulting shape is measured with the goal of inferring physical parameters such as elastic moduli [21]. Because these systems involve small length and time scales, measurements can be subject to substantial uncertainty, highlighting the need for accurate models that connect observed shapes with underlying mechanics.

A celebrated mathematical framework for predicting the shapes of elastic fluid interfaces is the Willmore energy and its extensions, which continues to find use as a model for a wide range of biological and soft-matter systems, including lipid bilayer membranes [11], giant unilamellar vesicles [14], colloidal membranes composed of rod-like viruses [1], and amphiphilic polymer layers [9].

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Despite the inherently three-dimensional nature of these systems, many theoretical studies focus on highly symmetric configurations, often because symmetry reduces the governing nonlinear partial differential equation (PDE) to more tractable forms. This simplification, however, can exclude physically important nonaxisymmetric configurations; indeed, recent studies have demonstrated that nonaxisymmetric effects can play a crucial role in shape transitions and mechanical responses that are more representative of realistic biological systems [12, 18]. Developing a tractable framework for nonaxisymmetric interfaces is therefore important both for understanding the broader range of physically accessible configurations and for connecting experimentally observed shapes to their underlying mechanical properties.

In this study, we develop such a framework to theoretically and computationally explore the mechanics of fluid layers with bending stiffness under compression and tension. In so doing, we dramatically generalize previous work such as [6] by discarding the restrictive assumption of axisymmetry, offering a way to analyze more complex shapes. Our article is structured as follows: first, we outline our problem and its relation to the classical minimal surface and buckling elastica problems, which provide useful intuition. We then thoroughly catalog the different shapes that can arise and compute the external axial force and torque, using both a perturbative approach and a pseudospectral solver for the full nonlinear problem. Our analysis is focused around two representative boundary configurations: the simplest case of parallel and concentric circles, which yields the catenoid in the Plateau–Douglas problem, and a pair of perturbed circles, which demonstrates the effects of more general, nonaxisymmetric boundaries. Both scenarios are found to yield physically relevant nonaxisymmetric shapes that potentially affect stability.

2 Background

We consider a surface $\mathcal{M}$ of prescribed area A whose energy E is given by the Willmore bending energy [20] plus an area term:

$$E = \frac{\kappa}{2} \int_{\mathcal{M}} dA (2H)^2 + \mu \int_{\mathcal{M}} dA, \quad (1)$$

where H is the mean curvature, κ is a bending modulus, dA is the area element, and μ is the tension, a Lagrange multiplier enforcing the constraint of fixed surface area. The Euler–Lagrange equation obtained by demanding the criticality of E is

$$\kappa(\Delta H - 2HK + 2H^3) = \mu H, \quad (2)$$

where K is the Gaussian curvature and Δ is the surface Laplace–Beltrami operator [17]. The associated boundary conditions are

$$\mathbf{X}|_{\partial\mathcal{M}} = \mathbf{Y} \quad \text{and} \quad \kappa(2H)|_{\partial\mathcal{M}} = 0, \quad (3)$$

where $\mathbf{X}$ is the surface position (to be determined) and $\mathbf{Y}$ is the position of the boundary (assumed to be given), which may consist of multiple components. Physically, Eqn. (3) represents the conditions of fluid–ring contact and vanishing of the bending moment at the boundary, respectively. Using standard cylindrical coordinates $\{\rho, \phi, z\}$, we consider surfaces of the form $\mathbf{X}(\phi, z) = (r(\phi, z) \cos \phi, r(\phi, z) \sin \phi, z)$, $|z| < h/2$, with boundaries $\mathbf{Y}(\phi) = (r^\pm(\phi) \cos \phi, r^\pm(\phi) \sin \phi, \pm h/2)$, as illustrated in Fig. 1a. While this parameterization is unsuitable for modeling highly buckled shapes for which r is not a function of z , it nonetheless suffices for shapes under small and intermediate compression. In this parameterization,

$$2H = \frac{1}{g^{3/2}} [r(r_z^2 + 1)(r_{\phi\phi} - r) - 2r_\phi^2 - 2rr_z r_\phi r_{z\phi} + rr_{zz}(r^2 + r_\phi^2)], \quad (4)$$

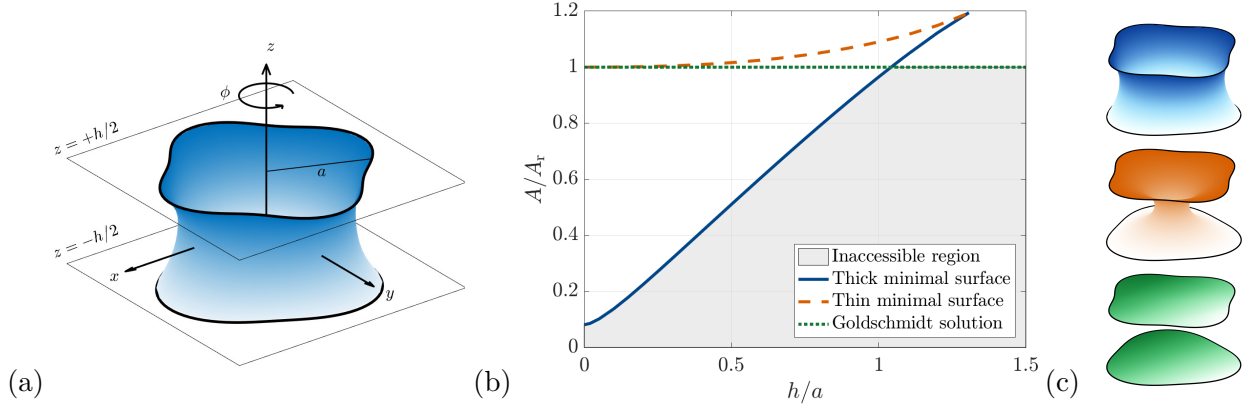


Figure 1: (a) A minimal surface connecting two parallel planar rings with separation $h = a$. The surface is parameterized by an axial coordinate z and azimuthal coordinate ϕ . The boundaries are $r_+(\phi) = a(1 + 0.1 \cos 4\phi)$, and $r_-(\phi) = a(1 + 0.1 \cos 3\phi)$. (b) Area versus extension for minimal surfaces with these boundary rings. No surfaces exist in the grey portion of the diagram. (c) The thick, thin, and Goldschmidt minimal surfaces for $h^* = a$.

$$K = \frac{1}{g^2} [rr_{zz}[r(r_{\phi\phi} - r) - 2r_\phi^2] - (rr_{z\phi} - r_z r_\phi)^2], \quad (5)$$

$$\Delta = \frac{1}{\sqrt{g}} \left[\frac{\partial}{\partial \phi} \left(\frac{1+r_z^2}{\sqrt{g}} \frac{\partial}{\partial \phi} - \frac{r_\phi r_z}{\sqrt{g}} \frac{\partial}{\partial z} \right) + \frac{\partial}{\partial z} \left(-\frac{r_\phi r_z}{\sqrt{g}} \frac{\partial}{\partial \phi} + \frac{r^2 + r_\phi^2}{\sqrt{g}} \frac{\partial}{\partial z} \right) \right], \quad (6)$$

where $r_z = \partial r / \partial z$, $r_\phi = \partial r / \partial \phi$, etc., and $g = r^2(1+r_z^2) + r_\phi^2$. Substituting Eqns. (4)–(6) into Eqn. (2) then yields a fourth-order PDE for the unknown function r . External forces and torques at the boundary rings are generally needed in order to impose Eqn. (3a). These quantities are necessarily constant, as follows from Noether's theorem: since the integrand of the energy functional Eqn. (1), $\mathcal{E} = [(\kappa/2)(2H)^2 + \mu]\sqrt{g}$, does not depend on z or ϕ explicitly, conserved quantities associated with these variables exist. We find that

$$F = \oint d\phi \left[\mathcal{E} - r_z \frac{\partial \mathcal{E}}{\partial r_z} - r_{zz} \frac{\partial \mathcal{E}}{\partial r_{zz}} + r_z \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right) - r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right] \quad (7)$$

is the conserved quantity associated with z and

$$T = - \oint d\phi \left[r_\phi \frac{\partial \mathcal{E}}{\partial r_z} + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{zz}} - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - r_\phi \frac{\partial}{\partial \phi} \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right) \right] \quad (8)$$

is the conserved quantity associated with ϕ [4]. Based on dimensions and the expressions in the axisymmetric limit, we might expect F to represent the axial force and T to represent the torque about the z -axis; the principle of virtual work can be used to confirm this (see Supplementary Material and also [19] for the $\kappa = 0$ case).

While the general solution to Eqn. (2) is complicated, we recognize that the two-boundary configuration studied here is very reminiscent of two classic problems in the calculus of variations, whose solutions offer valuable guiding intuition. Noting that the trivial solution $H = 0$ satisfies both the shape equation Eqn. (2) and the no-bending-moment boundary condition Eqn. (3b), and assuming that this shape is compatible with the fixed area constraint, we see that our problem becomes the problem of Plateau (or Douglas) [10]: what is the surface of minimal area that has a given boundary? The answer to this question is well known when the boundary consists of

parallel concentric circles of radius a : in this case, the only connected surfaces with $H = 0$ (i.e., minimal surfaces) are catenoids and the Goldschmidt solution. The former are surfaces of revolution with meridians of the form $r = b \cosh(z/b)$, $0 < b < 1$, while the latter is simply two discs connected center to center by a line. Catenoids cease to be a solution when $h/a > 1.3255\dots$, as the transcendental equation $b \cosh(h/(2b)) = a$, which encodes the boundary condition of fluid–boundary contact, has no solution for b . For h/a less than this value, there are actually two possible catenoids, a thick one (larger b) and a thin one (smaller b). Substituting $r = b \cosh(z/b)$ into Eqns. (7) and (8) shows that $F = 2\pi\mu b$ and $T = 0$ for catenoids.

Generalizations of the Plateau–Dogulas problem to noncircular boundaries have been studied [19, 2], and many properties of the axisymmetric case carry over. A plot of the solution area as a function of extension for a particular choice of noncircular boundary is shown in Fig. 1b, and visualizations of the three minimal surfaces for $h = a$, analogous to the two types of catenoid and Goldschmidt solution, are shown in Fig. 1c. In Fig. 1b, the solid blue line represents a family of relatively “thick” surfaces that are global minimizers of area. There is also a branch of “thin” solutions (dashed red line) that are theoretically feasible with the same h . However, these surfaces are not area-minimizing: such films are unstable to perturbations and thus do not occur in typical experiments. The last possible branch is the nonaxisymmetric Goldschmidt solution (dotted green line), whose area is A_r , the total area spanned by the two rings. Beyond the maximal extension for which the thick and thin branches intersect, this Goldschmidt solution is the only possible one.

Our problem can additionally be viewed as a higher-dimensional analogue of the classical elastica problem, in which an inextensible, one-dimensional rod buckles when compressed at its ends. This straight-to-buckled transition is the quintessential example of an elastic instability [5]. If the rod remains planar, its energy can be expressed as $(B/2) \int ds k^2 + \lambda \int ds$, where k is the curvature of the rod, B is its bending stiffness, ds is the arc length element, and λ is the tension that preserves total length. Note the resemblance between this energy and Eqn. (1): the first term is an integral of extrinsic curvature squared, and the second imposes a constant geometric measure. Analysis of the linearized Euler–Lagrange equation of the elastica bending energy reveals an eigenvalue problem with infinitely many sinusoidal eigenfunctions, each requiring a compressive force $F_n = B(n\pi/\varsigma)^2$, where $n = 1, 2, \dots$ is the mode number and ς is the rod length. In the same way, an elastic fluid interface sandwiched between two rings can be expected to exhibit infinitely many oscillatory modes, each requiring a compressive axial force.

3 Results: Circular Boundaries

We now consider the shape of a fixed-area fluid interface with out-of-plane bending resistance as the rings are pulled apart. In the phase diagram of Fig. 1b, this action corresponds to moving horizontally to the right. If $\bar{A} = A/A_r$ is less than or equal to 1, then moving to the right will eventually take one to the blue branch of thick minimal surfaces. The rings cannot be pulled any farther since this would entail entering the grey region of impossible surfaces with area less than that of a thick minimal surface. Therefore, our surfaces must have a maximum extension of $h^*(A)$, the extension corresponding to a thick minimal surface of area A . Furthermore, because $H = 0$ is a solution to the shape equation, the interface shape for $h = h^*$ will be precisely the minimal surface solution to the Plateau–Douglas problem.

It remains to calculate the force needed to hold these minimal surfaces in equilibrium. To do this, we expand upon an idea from [6] and define a dimensionless parameter $\epsilon = |h - h^*|/a$; this parameter is small when the fluid interface is stretched to nearly its maximal extension. We expand $r = r^{(0)} + \sqrt{\epsilon}r^{(1/2)} + \epsilon r^{(1)} + \dots$ and H , K , Δ and μ similarly. If the base quantity $r^{(0)}$ is a minimal

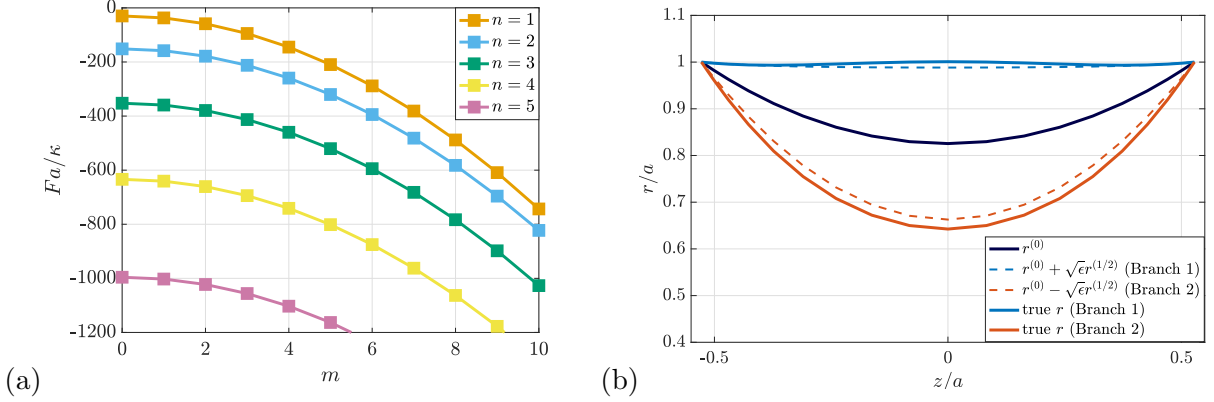


Figure 2: (a) Axial force for the catenoid ($\bar{A} = 1$) as a function of azimuthal mode number m for various axial mode numbers n . (b) Base (catenoid) and two area-preserving $O(\sqrt{\epsilon})$ perturbation solutions for $n = 1, m = 0$ ($h^*/a = 1.0554\dots$, $\epsilon = 0.05h^*/a$).

surface, using these representations and $H^{(0)} = 0$ in Eqns. (2) and (3b) yields the $O(\sqrt{\epsilon})$ eigenvalue problem

$$-\kappa \mathcal{J}^{(0)} H^{(1/2)} = \mu^{(0)} H^{(1/2)}, \quad H^{(1/2)}(\phi, z = \pm h/2) = 0 \quad (9)$$

for $H^{(1/2)}$. Here, $\mathcal{J}^{(0)} = -\Delta^{(0)} + 2K^{(0)}$ is the Jacobi operator of the base minimal surface; this eigenvalue problem is entirely analogous to the ones obtained when analyzing buckling instabilities of elastic rods, plates, and shells [8]. In the special case of a catenoid base state $r^{(0)} = b \cosh(z/b)$, $K^{(0)} = -2/\cosh^4(z/b)$, and $\Delta^{(0)} = \text{sech}^2(z/b)(\partial_z^2 + b^{-2}\partial_\phi^2)$. Separation of variables can then be used: letting $H^{(1)} = F_m(z)G_m(\phi)$, we find that $G_m(\phi) = A_m \cos m\phi + B_m \sin m\phi$, where A_m and B_m are constants, m is an integer, and $F_m(z)$ solves

$$\frac{1}{\cosh^2 \frac{z}{b}} \left(\partial_z^2 - \frac{m^2}{b^2} + \frac{2}{b^2 \cosh^2 \frac{z}{b}} \right) F_m = \frac{\mu^{(0)}}{\kappa} F_m, \quad F_m(z = \pm h/2) = 0. \quad (10)$$

The modes are thus characterized by ordered pairs of integers (n, m) , $n \geq 1, m \geq 0$. The permissible values of the tension (scaled by κ) are the eigenvalues of $-\mathcal{J}^{(0)}$; recalling that $F = 2\pi\mu b$ for a catenoid of neck b gives the external force for each mode, illustrated in Fig. 2a. When the boundary is circular, the cosine and sine eigenfunctions are simply rotations of each other; thus, the eigenvalues corresponding to a given n and m are doubly degenerate (except for $m = 0$, where the sine mode is spurious). We remark that while Eqn. (10) lacks a closed solution in terms of common functions, it closely resembles the associated Legendre equation after the transformation $\eta = \tanh(z/b)$, admitting a simple Frobenius series solution in η . However, we do not find this representation enlightening, so we will not pursue it here, relying instead on a numerical approach.

As an eigenfunction, $H^{(1/2)}$ is only determined up to a multiplicative constant, and thus so is $r^{(1/2)}$. To find this constant, we expand the area constraint:

$$0 = \sqrt{\epsilon} \oint d\phi \int_{-h/2}^{h/2} dz \sqrt{g^{(1/2)}} + \epsilon \left[-2 \sqrt{g^{(0)}} \Big|_{z=\frac{h}{2}} + \oint d\phi \int_{-h/2}^{h/2} dz \sqrt{g^{(1)}} \right]. \quad (11)$$

Expanding Eqn. (3a) to $O(\epsilon)$ yields the boundary conditions $r^{(0)} = a$, $r^{(1/2)} = 0$, and $r^{(1)} = \pm r_z^{(0)}$ at $z = \pm h/2$. Because the base solution is a minimal surface, the first variation of area vanishes in

the bulk, so the $O(\sqrt{\epsilon})$ z -integral reduces to boundary terms proportional to $r^{(1/2)}$ at $z = \pm h/2$. Since $r^{(1/2)} = 0$ at the boundary, the $O(\sqrt{\epsilon})$ term vanishes completely, and the area constraint is automatically satisfied at this order. We thus must examine the $O(\epsilon)$ terms to determine $H^{(1/2)}$. To this end, we compute $\sqrt{g}^{(0)}|_{z=h/2} = b \cosh^2(h/2b) = 1/b$ and

$$\sqrt{g}^{(1)} = \cosh \frac{z}{b} r^{(1)} + b \sinh \frac{z}{b} r_z^{(1)} + \frac{(r_\phi^{(1/2)})^2}{2b \cosh^2 \frac{z}{b}} + \tanh \frac{z}{b} r^{(1/2)} r_z^{(1/2)} + \frac{b(r_z^{(1/2)})^2}{2 \cosh^2 \frac{z}{b}}. \quad (12)$$

The $r^{(1)}$ terms again form a total derivative and their z -integral thus also reduces to boundary terms. As a result, Eqn. (11) becomes

$$4\pi b \sin^2 \frac{h}{2b} + \oint d\phi \int_{-h}^h dz \left[\frac{(r_\phi^{(1/2)})^2}{2b \cosh^2 \frac{z}{b}} + \tanh \frac{z}{b} r^{(1/2)} r_z^{(1/2)} + \frac{b(r_z^{(1/2)})^2}{2 \cosh^2 \frac{z}{b}} \right] = \frac{4\pi}{b}, \quad (13)$$

which determines the normalization constant. Notably, since $r^{(1/2)}$ and its derivatives appear in pairwise products, both negative and positive constants are valid. This suggests that there are two branches emanating from each eigensolution as h is reduced from h^* . Figure 2b shows these branches for the $(n, m) = (1, 0)$ catenoid of area $\bar{A} = 1$ and their approximations.

We now turn to the fully nonlinear problem, for which we use a pseudospectral method [16]. To conform with the traditional spectral formulation, we use a mixed Chebyshev–Fourier basis and treat both r and H as independent variables and formulate the fourth-order PDE as a pair of second-order PDEs, Eqns. (4) and (2), with respective Dirichlet boundary conditions, Eqns. (3a) and (3b). We also consider the tension μ as an unknown, balancing this additional variable with an equation for the surface area constraint. The discretized system of nonlinear equations is solved using a trust region algorithm with a stopping condition of the dimensionless ℓ^∞ residual being less than 10^{-10} . Figure 3 shows solutions for $\bar{A} = 1$ and $h/h^* = 0.9$ for the first few n and m modes, along with their corresponding eigensurfaces at $h = h^*$, found using the solver. As n and m increase, the surfaces exhibit more oscillations, developing a more pronounced checkerboard pattern. Consistent with the results of the perturbation theory, each (n, m) mode has two branches. The two branches consist of (sometimes subtly) distinct surfaces when n is odd, while the branches for modes of even n are mirror images of each other. The odd n modes are symmetric about the midplane, that is, they are even functions of z , but the modes of even n are not odd functions of z . With increasing m or n , the force F generally becomes more negative (compressive); a notable exception to this trend is $m = 1$, for which the forces required are smaller than for all other m . We attribute this to the $m = 1$ modes being essentially translations of cross-sections, involving comparatively little bending in the azimuthal direction and thus requiring less force, making the system more susceptible to a snap-through-type deformation [18]. All of the shapes in Fig. 3 require $T = 0$.

4 Results: Noncircular Boundaries

We now explore how noncircular boundaries affect the surface shape and ring force. To this end, we consider boundaries of the form $r^\pm(\phi)/a = 1 + \delta \sum_k e^{ik\phi} \hat{r}_k^\pm$, $0 < \delta \ll 1$, and perturbatively calculate the minimal surface, taken to be of the form $r^{[0]}(z) + \delta r^{[1]}(\phi, z) + \dots$, where $r^{[0]} = b \cosh(z/b)$ and $r^{[1]} = \sum_k e^{ik\phi} \hat{r}_k(z)$. Expanding Eqn. (4) to $O(\delta)$ gives a boundary value problem for the coefficients $\hat{r}_k(z)$ (see also [19]):

$$\frac{1}{2 \cosh^3 \frac{z}{b}} \left[\hat{r}_k'' - \frac{2}{b} \tanh \frac{z}{b} \hat{r}_k' + \frac{1 - k^2}{b^2} \hat{r}_k \right] = 0, \quad \hat{r}_k(z = \pm h/2) = \hat{r}_k^\pm \quad (14)$$

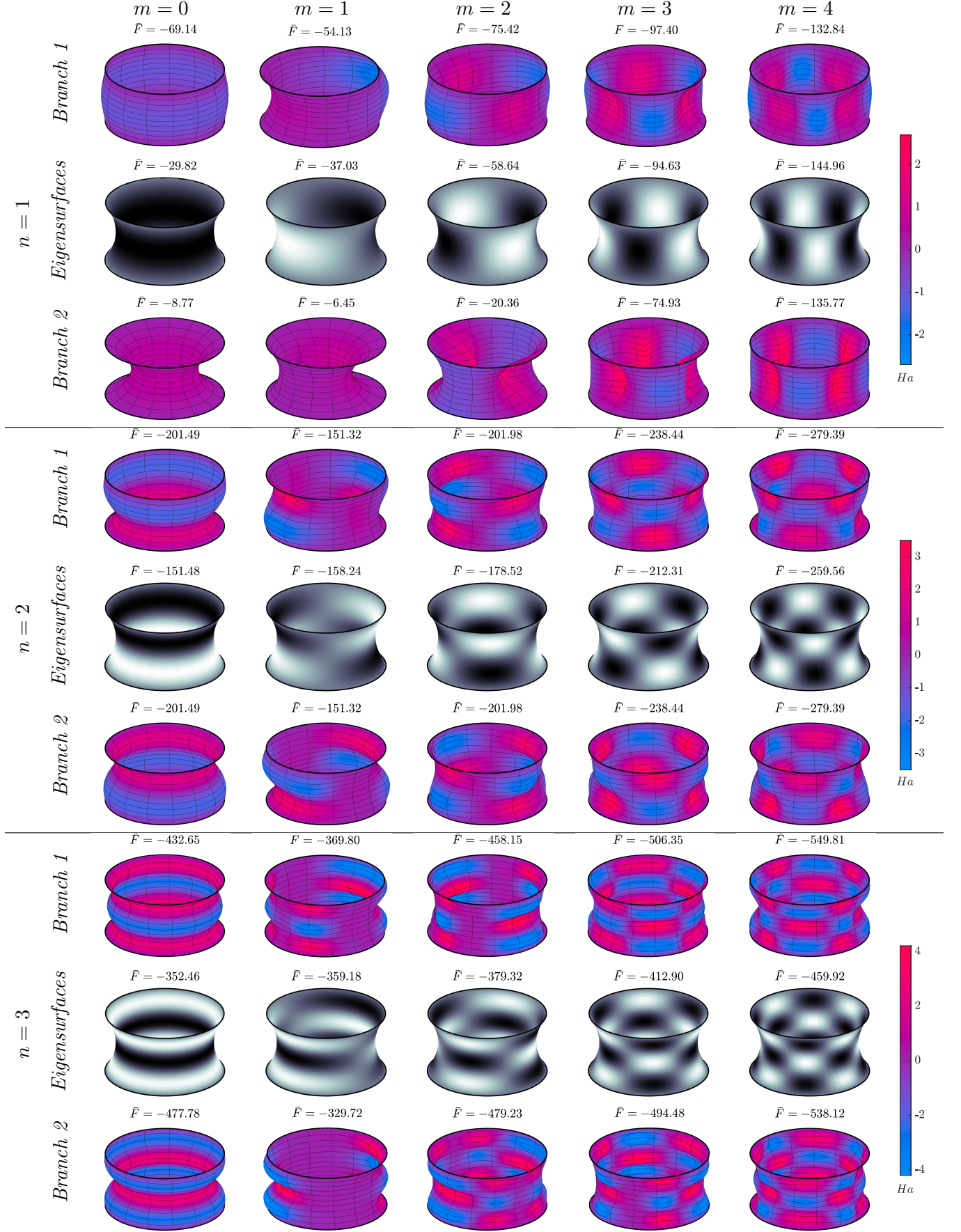


Figure 3: The eigensurfaces ($h = h^*$), corresponding buckling modes ($h/h^* = 0.9$), and dimensionless forces $\bar{F} = Fa/\kappa$, of an elastic fluid interface with circular rings ($\bar{A} = 1$).

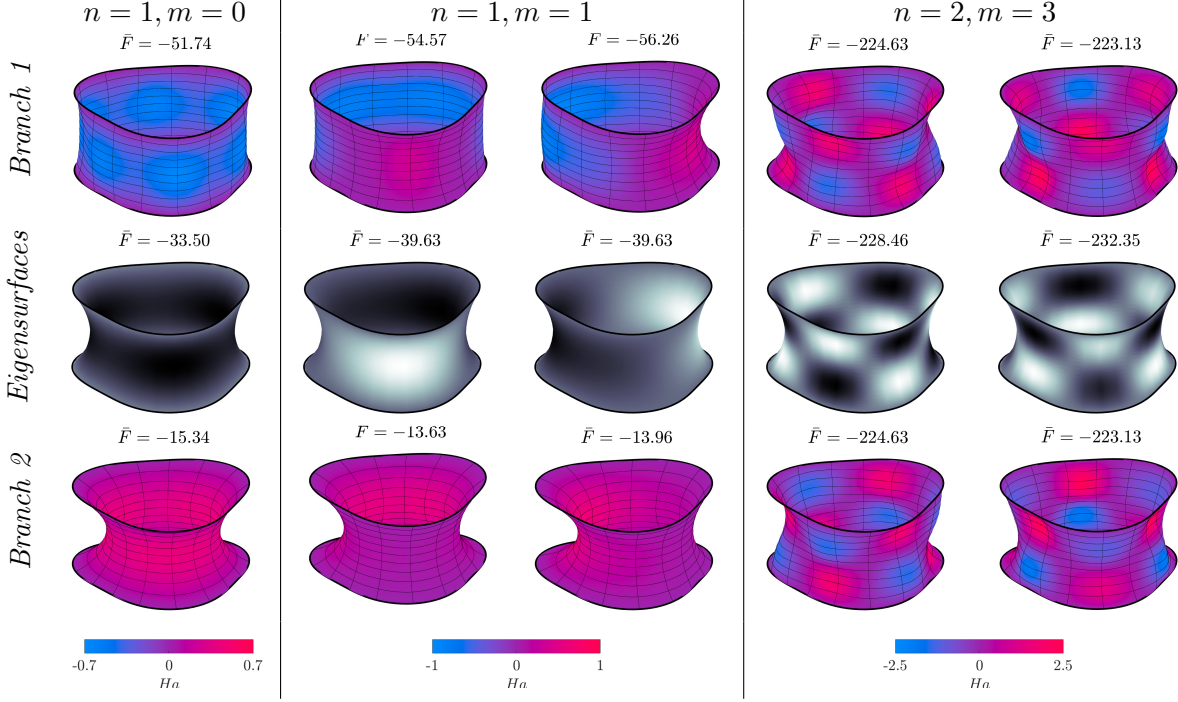


Figure 4: The eigensurfaces ($h = h^*$), corresponding buckling modes ($h/h^* = 0.9475\dots$), and dimensionless forces $\bar{F} = Fa/\kappa$, of an elastic fluid interface with boundary rings of the form $r^\pm(\phi)/a = 1 + 0.1 \cos(3\phi)$ ($\bar{A} = 1.0341\dots$). Sine and cosine modes are distinct.

We assume without loss that $\hat{r}_0^\pm = 0$ so that the area constraint is automatically satisfied at $O(\delta)$. The general solution of Eqn. (14) can be written using hypergeometric functions; using these $\hat{r}_k(z)$, the $O(\delta)$ Laplace–Beltrami operator and Gaussian curvature are

$$\Delta^{[1]} = -\frac{1}{\cosh^3 \frac{z}{b}} \sum_k e^{ik\phi} \left[\frac{2ik\hat{r}_k}{b^3 \cosh^2 \frac{z}{b}} \partial_\phi + \tanh \frac{z}{b} \left(\frac{(1-k^2)\hat{r}_k}{b^2} - \frac{2 \tanh \frac{z}{b} \hat{r}'_k}{b} + \hat{r}''_k \right) \partial_z \right. \\ \left. + \frac{2\hat{r}_k}{b^3} \partial_\phi^2 + \frac{2ik\hat{r}_k \tanh \frac{z}{b}}{b^2} \partial_z \partial_\phi + 2 \tanh \frac{z}{b} \hat{r}'_k \partial_z^2 \right] \quad (15)$$

$$K^{[1]} = -\frac{1}{b \cosh^5 \frac{z}{b}} \sum_k e^{ik\phi} \left[\hat{r}''_k - \frac{4}{b} \tanh \frac{z}{b} \hat{r}'_k - \frac{1-k^2}{b^2} \hat{r}_k \right], \quad (16)$$

from which we define $-\mathcal{J}^{[1]} = \Delta^{[1]} - 2K^{[1]}$. Classical perturbation theory for degenerate self-adjoint operators gives formulas for the corrections to the eigenvalues [3]: if $\{u_1^{[0]}, u_2^{[0]}\}$ are two orthonormal eigenfunctions of $-\mathcal{J}^{[0]}$ for a given mode (n, m) , $m > 0$, then the perturbed tensions $\mu_{nm}^{[1]}/\kappa$ are eigenvalues of the 2×2 matrix $M_{ij} = \langle -\mathcal{J}^{[1]} u_i^{[0]}, u_j^{[0]} \rangle$, where angular brackets denote the inner product. Clearly, only eigenvalues for which $r_k \neq 0$ will receive a nonzero perturbation, lifting the degeneracy at this order. Proceeding to $O(\delta^2)$ reveals that all eigenvalues for modes for which m is a multiple of a k with $r_k \neq 0$ are perturbed. The argument from Sec. 3 continues to hold, and we again have two branches per eigenfunction; however, this time $\cos(m\phi)$ and $\sin(m\phi)$ are not the same because of the perturbed boundary. We thus expect a total of *four* branches per (n, m) pair in the $m \neq 0$ nonaxisymmetric case. Examples of these surfaces for $(1, 0)$, $(1, 1)$, and $(2, 3)$ modes with a single-wavenumber boundary perturbation are shown in Fig. 4. We again see that some $m = 1$ modes require less compression than $m = 0$.

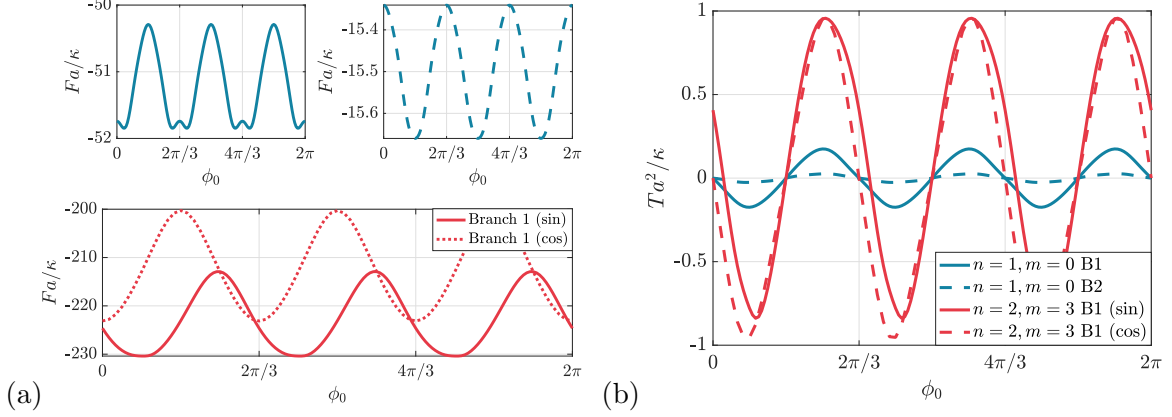


Figure 5: Force and torque as functions of relative ring twist ϕ_0 for some of the surfaces in Fig. 4. (a) Axial force for (1,0) mode branch 1 [top left], (1,0) mode branch 2 [top right], and the (2,3) mode branches [bottom]. (b) Torque for the branches in (a). B means branch.

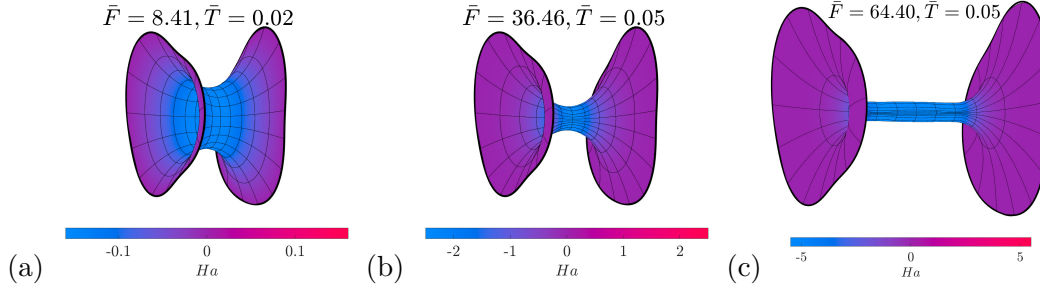


Figure 6: Transition of a (1,0) mode into a tether with no maximal extension ($\bar{A} = 1.1823\dots$). Boundary rings are $r^-(\phi)/a = 1 + 0.1(\cos \phi + \sin 2\phi + \cos 3\phi)$ and $r^+(\phi) = r^-(\phi - \pi/3)$. (a) $h/a = 1$, (b) $h/a = 1.5$, (c) $h/a = 3$.

Noncircular boundary rings allow for the possibility of a torque, Eqn. (8), due to rotational symmetry breaking. A boundary perturbation also introduces periodic $O(\delta)$ variations in the force. Force and torque as functions of relative twist ϕ_0 for nonaxisymmetric boundary rings are plotted in Fig. 5. We find that as n increases, so does T . For $m > 0$, sin and cos branches tend to require roughly the same magnitude of force and torque. For completeness, we also explore the case of $\bar{A} > 1$. Here, the (1,0) branch of the thin minimal surface no longer has a maximal extension. For $h > h^*$, the surface becomes an elongated cylindrical tether that resembles a regularized Goldschmidt solution (Fig. 6) connecting two halves of a minimal surface. It is natural that tethers only emerge from this mode, as increasing h requires a positive (extensile) force, suppressing the buckling instability. As in the axisymmetric case [13], the cross sections of the tethers are nearly circular so that $\bar{T} = T a^2 / \kappa \approx 0$. The force $\bar{F}$, by contrast, is large, increasing linearly with extension for $h \gg a$.

5 Conclusion

Roughly speaking, fixed-area elastic fluid interfaces can be understood as amalgamations of classical minimal surfaces and elastic shells. For sufficiently small surface area, such interfaces have a maximal extension h^* and form a minimal surface as in the Plateau–Douglas problem. External

confining forces, in conjunction with the area constraint, introduce the possibility of buckled solutions, analogous to how compressing the ends of a slender inextensible elastic rod gives rise to buckling. Each of the infinitely many buckling modes characterized by integers n and m leads to multiple feasible branches as h is compressed, the precise number of which depends on the boundary symmetry. Interestingly, the $m = 1$ modes can require less force than $m = 0$, suggesting that the activation of these nonaxisymmetric modes could lower the energy barrier to shape transitions.

The work presented here broadens our understanding of feasible curved fluid configurations, providing a foundation for understanding biological processes involving asymmetry. Future work includes using a more general parameterization so that highly buckled shapes that are not functions of z can be modeled. Such a parameterization could in principle also be used to treat nonplanar and deformable boundaries. A more general energy that includes a Gaussian curvature term also offers an interesting system to investigate, as Eqn. (3) becomes inhomogeneous, prohibiting $H = 0$. Singular perturbation theory will thus be needed.

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Supplementary Material for “Shapes, forces, and torques of compressed elastic fluid interfaces: beyond axisymmetric configurations”

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1 Geometric quantities

Let the surface be given by

$$\mathbf{X}(\phi, z) = (r(\phi, z) \cos \phi, r(\phi, z) \sin \phi, z) \quad (1)$$

Tangent vectors:

$$\mathbf{X}_\phi = (r_\phi \cos \phi - r \sin \phi, r_\phi \sin \phi + r \cos \phi, 0) \quad (2)$$

$$\mathbf{X}_z = (r_z \cos \phi, r_z \sin \phi, 1) \quad (3)$$

Metric tensor $g_{ij} = \mathbf{X}_i \cdot \mathbf{X}_j$:

$$g_{ij} = \begin{bmatrix} r^2 + r_\phi^2 & r_\phi r_z \\ r_\phi r_z & r_z^2 + 1 \end{bmatrix} \quad (4)$$

Also, $g = \det(g_{ij}) = r^2(r_z^2 + 1) + r_\phi^2$.

Normal vector:

$$\mathbf{N} = \frac{(r_\phi \sin \phi + r \cos \phi, r \sin \phi - r_\phi \cos \phi, -rr_z)}{\sqrt{g}} \quad (5)$$

Second derivatives:

$$\mathbf{X}_{\phi\phi} = (r_{\phi\phi} \cos \phi - 2r_\phi \sin \phi - r \cos \phi, r_{\phi\phi} \sin \phi + 2r_\phi \cos \phi - r \sin \phi, 0) \quad (6)$$

$$\mathbf{X}_{\phi z} = (r_{z\phi} \cos \phi - r_z \sin \phi, r_{\phi z} \sin \phi + r_z \cos \phi, 0) \quad (7)$$

$$\mathbf{X}_{zz} = (r_{zz} \cos \phi, r_{zz} \sin \phi, 0) \quad (8)$$

Curvature tensor $b_{ij} = -\mathbf{N} \cdot \mathbf{X}_{ij}$:

$$b_{ij} = -\frac{1}{\sqrt{g}} \begin{bmatrix} r(r_{\phi\phi} - r) - 2r_\phi^2 & rr_{z\phi} - r_z r_\phi \\ rr_{z\phi} - r_z r_\phi & rr_{zz} \end{bmatrix} \quad (9)$$

Also, $b = \det(b_{ij}) = (rr_{zz}[r(r_{\phi\phi} - r) - 2r_\phi^2] - (rr_{z\phi} - r_z r_\phi)^2)/g$.

Mean and Gaussian curvatures:

$$2H = g^{ij}b_{ij} = \frac{r(r_z^2 + 1)(r_{\phi\phi} - r) - 2r_\phi^2 - 2rr_z r_\phi r_{z\phi} + rr_{zz}(r^2 + r_\phi^2)}{[r^2(r_z^2 + 1) + r_\phi^2]^{3/2}} \quad (10)$$

$$K = \frac{b}{g} = \frac{rr_{zz}[r(r_{\phi\phi} - r) - 2r_\phi^2] - (rr_{z\phi} - r_z r_\phi)^2}{[r^2(r_z^2 + 1) + r_\phi^2]^2} \quad (11)$$

respectively.

2 Variational Formulation

2.1 Noether invariants

Let $\mathcal{E}(r, r_z, r_\phi, r_{zz}, r_{z\phi}, r_{\phi\phi})$ be defined by $E = \int d\phi dz \mathcal{E}$. Note that $\mathcal{E}$ includes the $\sqrt{g}$ factor and thus has dimensions of energy per unit length (or force). Since $\mathcal{E}$ does not depend on z or ϕ explicitly, we have by the chain rule

$$\frac{\partial \mathcal{E}}{\partial z} = \frac{\partial \mathcal{E}}{\partial r} r_z + \frac{\partial \mathcal{E}}{\partial r_z} r_{zz} + \frac{\partial \mathcal{E}}{\partial r_\phi} r_{z\phi} + \frac{\partial \mathcal{E}}{\partial r_{zz}} r_{zzz} + \frac{\partial \mathcal{E}}{\partial r_{z\phi}} r_{z\phi z} + \frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} r_{\phi\phi z} \quad (12)$$

$$\frac{\partial \mathcal{E}}{\partial \phi} = \frac{\partial \mathcal{E}}{\partial r} r_\phi + \frac{\partial \mathcal{E}}{\partial r_z} r_{z\phi} + \frac{\partial \mathcal{E}}{\partial r_\phi} r_{\phi\phi} + \frac{\partial \mathcal{E}}{\partial r_{zz}} r_{zz\phi} + \frac{\partial \mathcal{E}}{\partial r_{z\phi}} r_{z\phi\phi} + \frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} r_{\phi\phi\phi} \quad (13)$$

Also, we have the Euler equation

$$\frac{\partial \mathcal{E}}{\partial r} - \left(\frac{\partial \mathcal{E}}{\partial r_z} \right)_z - \left(\frac{\partial \mathcal{E}}{\partial r_\phi} \right)_\phi + \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_{zz} + \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_{z\phi} + \left(\frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} \right)_{\phi\phi} = 0 \quad (14)$$

which we use to eliminate $\partial \mathcal{E} / \partial r$. After some simplifying and rewriting of derivatives, we find

$$\mathcal{E}_z = \left[r_z \frac{\partial \mathcal{E}}{\partial r_z} + r_{zz} \frac{\partial \mathcal{E}}{\partial r_{zz}} - r_z \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right]_z + \left[r_z \frac{\partial \mathcal{E}}{\partial r_\phi} - r_z \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_z + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} - r_z \left(\frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} \right)_\phi \right]_\phi \quad (15)$$

$$\mathcal{E}_\phi = \left[r_\phi \frac{\partial \mathcal{E}}{\partial r_z} + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{zz}} - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_\phi \right]_z + \left[r_\phi \frac{\partial \mathcal{E}}{\partial r_\phi} - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_z + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{z\phi}} - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} \right)_\phi \right]_\phi \quad (16)$$

In order to find the conserved quantities, we integrate both expressions in ϕ so that the ϕ -derivative terms vanish. We thus conclude that

$$F := \oint d\phi \left[\mathcal{E} - r_z \frac{\partial \mathcal{E}}{\partial r_z} - r_{zz} \frac{\partial \mathcal{E}}{\partial r_{zz}} + r_z \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right] \quad (17)$$

$$T := - \oint d\phi \left[r_\phi \frac{\partial \mathcal{E}}{\partial r_z} + r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{zz}} - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - r_\phi \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_\phi \right] \quad (18)$$

are constant. Based on dimensions ($\mathcal{E}$ has dimensions of energy per length, i.e. force) and the expressions in the $r_\phi \rightarrow 0$ limit, we might expect F to represent the axial force and T to represent the torque (twisting moment) about the z -axis. We will show this below.

The full expressions for F and T are complicated, so we will calculate them numerically. To this end, we calculate

$$\frac{\partial \mathcal{E}}{\partial r_{zz}} = \sqrt{g} \left[\kappa(2H) \frac{\partial(2H)}{\partial r_{zz}} \right] = \kappa(2H) \frac{r(r^2 + r_\phi^2)}{g} \quad (19)$$

$$\frac{\partial \mathcal{E}}{\partial r_{z\phi}} = \sqrt{g} \left[\kappa(2H) \frac{\partial(2H)}{\partial r_{z\phi}} \right] = -\kappa(2H) \frac{2rr_z r_\phi}{g} \quad (20)$$

$$\frac{\partial \mathcal{E}}{\partial r_z} = \left[\kappa(2H) \frac{\partial(2H)}{\partial r_z} \right] \sqrt{g} + \frac{r^2 r_z}{\sqrt{g}} \left[\frac{\kappa}{2} (2H)^2 + \mu \right] \quad (21)$$

$$\frac{\partial(2H)}{\partial r_z} = \frac{2rr_z(r_{\phi\phi} - r) - 2rr_\phi r_{z\phi}}{g^{3/2}} - \frac{3}{2}(2H) \frac{2r^2 r_z}{g} \quad (22)$$

$$\frac{\partial K}{\partial r_z} = \frac{2r_\phi(rr_{z\phi} - r_z r_\phi)}{g^2} - 2K \frac{2r^2 r_z}{g} \quad (23)$$

2.2 Principle of virtual work

For simplicity, we take the left ring to be fixed and move the right one. We will show that the conserved quantities above correspond to axial force and torque, using an argument that generalizes Jia et al. (2021). By the principle of virtual work

$$\delta E - \oint d\ell f_1 \delta z_1 - \oint d\ell \tau_1 \delta \phi_1 - \oint d\ell m_1 \delta \theta_1 = 0 \quad (24)$$

Here, $d\ell$ is the boundary line element, f_1 is the axial force per unit length on the boundary, τ_1 is the twisting moment per unit length on the boundary, m_1 is the bending moment per unit length on the boundary, δz_1 is the virtual displacement of the ring, $\delta \phi_1$ is the virtual change in azimuthal angle by which the ring is rotated, $\delta \theta_1$ is the virtual change in the angle at the boundary (where $\tan \theta_1(\phi) = r_z(z_1, \phi)$), and the integrals are taken with respect to boundary arc length. In particular,

$$d\ell = d\phi \sqrt{\bar{r}^2 + \bar{r}_\phi^2} \quad (25)$$

where $\bar{r}(\phi)$ is the curve representing the boundary ring.

Next, we calculate the first variation. Let $\mathcal{E}$ be the energy density as above.

$$\begin{aligned} \delta E &= \int_0^{2\pi} d\phi \int_{-z_1}^{z_1} dz [\mathcal{E}(r + \delta r, \dots, r_{\phi\phi} + \delta r_{\phi\phi}) - \mathcal{E}(r, \dots, r_{\phi\phi})] + \oint d\phi \delta z_1 \mathcal{E} \\ &= \int_0^{2\pi} d\phi \int_{-z_1}^{z_1} dz \left[\delta r \frac{\partial \mathcal{E}}{\partial r} + \dots + \delta r_{\phi\phi} \frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} \right] + \oint d\phi \delta z_1 \mathcal{E} \\ &= \int_0^{2\pi} d\phi \int_{-z_1}^{z_1} dz \delta r \frac{\delta \mathcal{E}}{\delta r} + \oint d\phi \left\{ \delta z_1 \mathcal{E} + \delta r \left[\frac{\partial \mathcal{E}}{\partial r_z} - \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_\phi \right] + \delta r_z \frac{\partial \mathcal{E}}{\partial r_{zz}} \right\} \end{aligned} \quad (26)$$

where

$$\frac{\delta \mathcal{E}}{\delta r} = \frac{\partial \mathcal{E}}{\partial r} - \left(\frac{\partial \mathcal{E}}{\partial r_z} \right)_z - \left(\frac{\partial \mathcal{E}}{\partial r_\phi} \right)_\phi + \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_{zz} + \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_{z\phi} + \left(\frac{\partial \mathcal{E}}{\partial r_{\phi\phi}} \right)_{\phi\phi} \quad (27)$$

is assumed to be zero in the Euler–Lagrange equation. Next, we need to write δr and δr_z in terms of other variations. Let δr_1 represent the difference in r for the point at the boundary (i.e., the point whose Lagrangian label is $t = 0$).

$$\delta r_1 = (r + \delta r)(z_1 + \delta z_1, \phi_1 + \delta \phi_1) - r(z_1, \phi_1) = r_z(z_1, \phi_1) \delta z_1 + r_\phi(z_1, \phi_1) \delta \phi_1 + \delta r(z_1, \phi_1) + \dots \quad (28)$$

$$\implies \delta r(z_1, \phi_1) = \delta r_1 - r_z(z_1, \phi_1) \delta z_1 - r_\phi(z_1, \phi_1) \delta \phi_1 \quad (29)$$

What is the right formula for the bending moment when the axis and force aren't orthogonal?

Let $\mathbf{e}_1$ be the tangent vector of the boundary curve and $\mathbf{e}_3$ be the surface normal. Then, $\mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_1$ is tangent to the surface. The angle θ_1 is the angle between $\mathbf{e}_2$ and the unit vector in the z -direction. We find that

$$\cos \theta_1 = \mathbf{e}_2 \cdot \mathbf{e}_z|_{z=z_1, \phi=\phi_1} = \frac{\sqrt{r^2 + r_\phi^2}}{\sqrt{r^2(1 + r_z^2) + r_\phi^2}} \Big|_{z=z_1, \phi=\phi_1} \quad (30)$$

Alternatively, we could work with $\tan \theta_1$ to be consistent with Jia et al. (2021). In this case, we will want to vary

$$\tan \theta_1 = \frac{rr_z}{\sqrt{r^2 + r_\phi^2}} \Big|_{z=z_1, \phi=\phi_1} \quad (31)$$

We perturb $\phi_1 \rightarrow \phi_1 + \delta \phi_1$, $z \rightarrow z + \delta z_1$, $r \rightarrow r + \delta r$. We also use the expression for δr that we derived above to eliminate it in favor of δr_1 , δz_1 , and $\delta \phi_1$. The computation is lengthy, but the result is

$$\begin{aligned} \delta \theta_1 \sec^2 \theta_1 &= \delta \theta_1 \frac{g}{r^2 + r_\phi^2} = \frac{r \delta z_1}{\sqrt{r^2 + r_\phi^2}} \left(r_{zz} - \frac{r_z r_\phi r_{z\phi}}{r^2 + r_\phi^2} \right) + \frac{r \delta \phi_1}{\sqrt{r^2 + r_\phi^2}} \left(r_{z\phi} - \frac{r_z r_\phi r_{\phi\phi}}{r^2 + r_\phi^2} \right) \\ &\quad + \frac{\delta r_1}{\sqrt{r^2 + r_\phi^2}} \left(r_z - \frac{r^2 r_z}{r^2 + r_\phi^2} \right) - \frac{\delta r_\phi}{\sqrt{r^2 + r_\phi^2}} \frac{r r_z r_\phi}{r^2 + r_\phi^2} + \frac{r \delta r_z}{\sqrt{r^2 + r_\phi^2}} \end{aligned} \quad (32)$$

We want to solve for δr_z , so we rearrange to get

$$\delta r_z = \delta\theta_1 \frac{g}{r\sqrt{r^2 + r_\phi^2}} - \delta z_1 \left(r_{zz} - \frac{r_z r_\phi r_{z\phi}}{r^2 + r_\phi^2} \right) - \delta\phi_1 \left(r_{z\phi} - \frac{r_z r_\phi r_{\phi\phi}}{r^2 + r_\phi^2} \right) - \delta r_1 \frac{r_z r_\phi^2}{r(r^2 + r_\phi^2)} + \delta r_\phi \frac{r_z r_\phi}{r^2 + r_\phi^2} \quad (33)$$

We use the fact that $\delta r_\phi = (\delta r_1)_\phi - r_{z\phi} \delta z_1 - r_{\phi\phi} \delta\phi_1$ (for our setup, δz_1 and $\delta\phi_1$ are constants). Plugging this into the boundary integral yields

$$\oint d\phi \delta r_z \frac{\partial \mathcal{E}}{\partial r_{zz}} = \oint d\phi \left[\delta\theta_1 \frac{g}{r\sqrt{r^2 + r_\phi^2}} - \delta z_1 r_{zz} - \delta\phi_1 r_{z\phi} \right] \frac{\partial \mathcal{E}}{\partial r_{zz}} - \oint d\phi \frac{\delta r_1}{r} \left(\frac{r r_z r_\phi}{r^2 + r_\phi^2} \frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_\phi \quad (34)$$

The rest of the terms are seen to cancel out. This equation implies that if $\bar{r}(\phi)$ and δr_1 are not constant, then the radial force will need to be modified by an extra term. However, in our setup, the rings are rigid so $\delta r_1 = 0$.

Adding in the external force, torque, and moment via the principle of virtual work, we find

$$\oint d\phi \left\{ \mathcal{E} - r_z \left[\frac{\partial \mathcal{E}}{\partial r_z} - \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_\phi \right] - r_{zz} \frac{\partial \mathcal{E}}{\partial r_{zz}} - f_1 \sqrt{r^2 + r_\phi^2} \right\} \delta z_1 = 0 \quad (35)$$

$$\oint d\phi \left\{ -r_\phi \left[\frac{\partial \mathcal{E}}{\partial r_z} - \left(\frac{\partial \mathcal{E}}{\partial r_{zz}} \right)_z - \left(\frac{\partial \mathcal{E}}{\partial r_{z\phi}} \right)_\phi \right] - r_{z\phi} \frac{\partial \mathcal{E}}{\partial r_{zz}} - \tau_1 \sqrt{r^2 + r_\phi^2} \right\} \delta\phi_1 = 0 \quad (36)$$

$$\oint d\phi \left\{ \frac{g}{r\sqrt{r^2 + r_\phi^2}} \frac{\partial \mathcal{E}}{\partial r_{zz}} - m_1 \sqrt{r^2 + r_\phi^2} \right\} \delta\theta_1 = 0 \quad (37)$$

for arbitrary deformations at the boundary. Setting the integrands to zero yields the expressions given in the previous section after integration by parts is performed. In particular, one can calculate that

$$\frac{\partial \mathcal{E}}{\partial r_{zz}} = \frac{r(r^2 + r_\phi^2)}{g} [\kappa(2H)] \quad (38)$$

for our energy. This yields the expression for the bending moment (per unit length) that we are familiar with: $m_1 = \kappa(2H)$.

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